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Task 1: Import NumPy, Pandas, Matplotlib.pyplot and Seaborn.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Task 2: Load train.csv into a Pandas DataFrame using pd.read_csv().

```
from google.colab import files
uploaded = files.upload()
```

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Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to

enable.

Task 3: Display first 10 rows using df.head(10).

```
df_train = pd.read_csv('Titanic-Dataset.csv')
display(df_train.head(10))
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...)	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S
5	6	0	3	Moran, Mr. James	male	NaN	0	0	330877	8.4583	NaN	Q
6	7	0	1	McCarthy, Mr. Timothy J	male	54.0	0	0	17463	51.8625	E46	S
7	8	0	3	Palsson, Master. Gosta Leonard	male	2.0	3	1	349909	21.0750	NaN	S

Task 4: Display last 5 rows using df.tail(5).

```
display(df_train.tail(5))
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.00	NaN	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.00	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.45	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.00	C148	C

Task 5: Check dataset dimensions using df.shape (Expected: 891 x 12).

```
print(df_train.shape)
```

(891, 12)

Task 6: Display dataset information using df.info().

```
df_train.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
```

```

4 Sex      891 non-null object
5 Age      714 non-null float64
6 SibSp    891 non-null int64
7 Parch    891 non-null int64
8 Ticket   891 non-null object
9 Fare     891 non-null float64
10 Cabin   204 non-null object
11 Embarked 889 non-null object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB

```

Task 7: Generate summary statistics using `df.describe()`; record mean, minimum and maximum values of Age and Fare.

```

summary_stats = df_train.describe()
display(summary_stats)

print(f"\nMean Age: {summary_stats.loc['mean', 'Age']:.2f}")
print(f"Minimum Age: {summary_stats.loc['min', 'Age']:.2f}")
print(f"Maximum Age: {summary_stats.loc['max', 'Age']:.2f}")

print(f"\nMean Fare: {summary_stats.loc['mean', 'Fare']:.2f}")
print(f"Minimum Fare: {summary_stats.loc['min', 'Fare']:.2f}")
print(f"Maximum Fare: {summary_stats.loc['max', 'Fare']:.2f}")

```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

Mean Age: 29.70
Minimum Age: 0.42
Maximum Age: 80.00

Mean Fare: 32.20
Minimum Fare: 0.00
Maximum Fare: 512.33

Task 8: Identify missing values using `df.isnull().sum()`.

```

missing_values = df_train.isnull().sum()
display(missing_values)

```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

Task 9: Fill missing Age values with the median.

```

median_age = df_train['Age'].median()
df_train['Age'].fillna(median_age, inplace=True)

```

```
# Verify that missing 'Age' values have been filled
print(f"Missing 'Age' values after filling: {df_train['Age'].isnull().sum()}")
```

Missing 'Age' values after filling: 0
 /tmp/ipykernel_420/3343285883.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment. The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting is a view.
 For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value, inplace=True)

```
df_train['Age'].fillna(median_age, inplace=True)
```

Task 10: Filter surviving passengers (Survived == 1).

```
df_survived = df_train[df_train['Survived'] == 1]
display(df_survived.head())
```

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	2	1	Cumings, Mrs. John Bradley (Florence Briggs Th...)	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S

Task 11: Convert Fare column to a NumPy array.

```
fare_array = df_train['Fare'].to_numpy()

print(f"Type of fare_array: {type(fare_array)}")
print(f"First 5 elements of fare_array: {fare_array[:5]}")
```

Type of fare_array: <class 'numpy.ndarray'>
 First 5 elements of fare_array: [7.25 71.2833 7.925 53.1 8.05]

Task 12: Calculate maximum and average Fare using np.max() and np.mean().

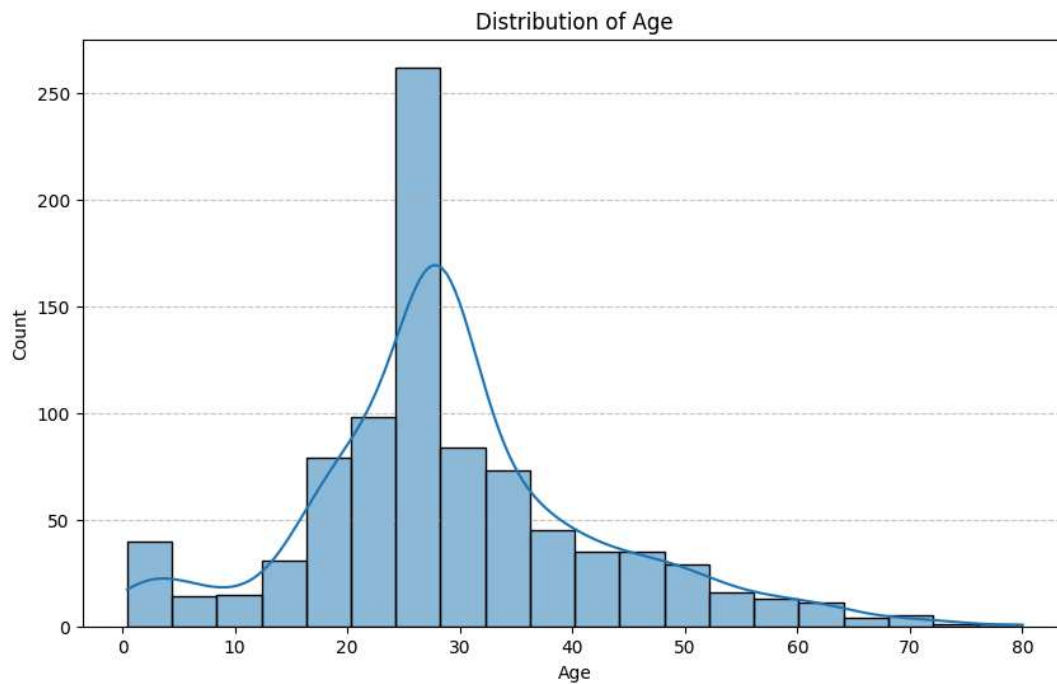
```
max_fare = np.max(fare_array)
mean_fare = np.mean(fare_array)

print(f"Maximum Fare: {max_fare:.2f}")
print(f"Average Fare: {mean_fare:.2f}")
```

Maximum Fare: 512.33
 Average Fare: 32.20

Task 13: Plot histogram of Age.

```
plt.figure(figsize=(10, 6))
sns.histplot(df_train['Age'], bins=20, kde=True)
plt.title('Distribution of Age')
plt.xlabel('Age')
plt.ylabel('Count')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```



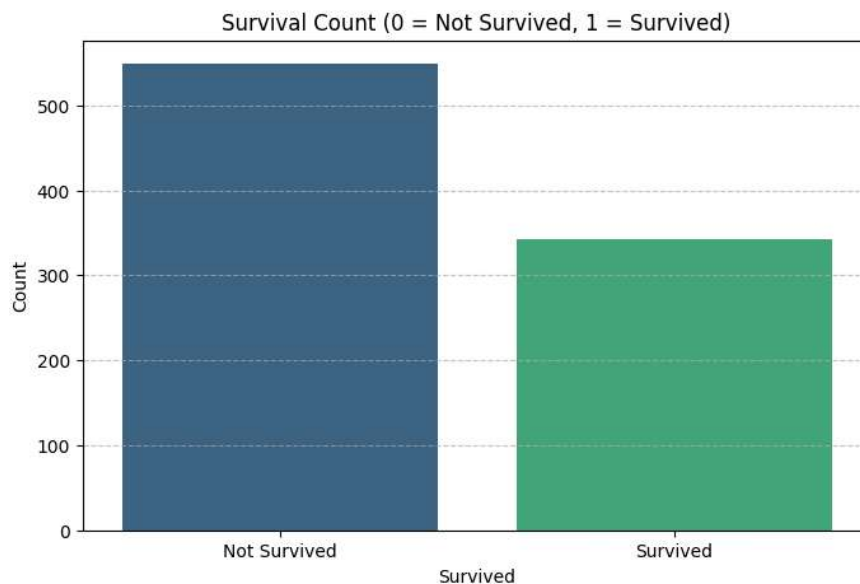
Task 14: Plot bar chart of Survival Count.

```
plt.figure(figsize=(8, 5))
sns.countplot(x='Survived', data=df_train, palette='viridis')
plt.title('Survival Count (0 = Not Survived, 1 = Survived)')
plt.xlabel('Survived')
plt.ylabel('Count')
plt.xticks(ticks=[0, 1], labels=['Not Survived', 'Survived'])
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```

/tmp/ipykernel_420/913352082.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `1`

```
sns.countplot(x='Survived', data=df_train, palette='viridis')
```



Task 15: Plot scatter graph of Age vs Fare.

```
plt.figure(figsize=(10, 6))
sns.scatterplot(x='Age', y='Fare', data=df_train, alpha=0.7)
plt.title('Scatter Plot of Age vs. Fare')
plt.xlabel('Age')
plt.ylabel('Fare')
plt.grid(True, linestyle='--', alpha=0.7)
plt.show()
```

Scatter Plot of Age vs. Fare

